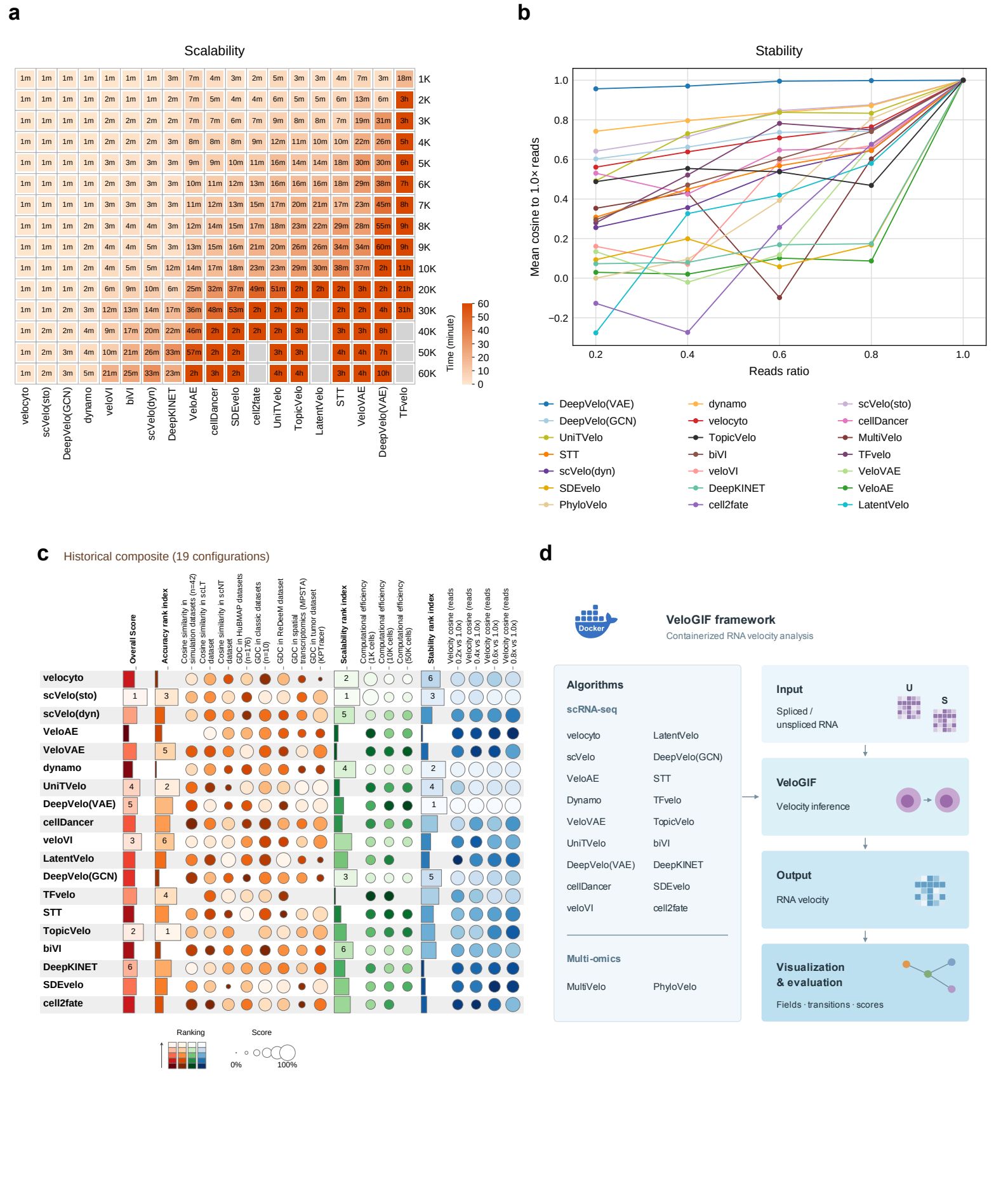

a
Scalability
1K 2K 3K 4K 5K 6K 7K 8K 9K 10K 20K 30K 40K 50K 60K
velocyto scVelo(sto) DeepVelo(GCN) dynamo veloVI biVI scVelo(dyn) DeepKINET VeloAE cellDancer SDEvelo cell2fate UniTVelo TopicVelo LatentVelo STT VeloVAE DeepVelo(VAE) TFvelo
Time (minute)
b
Stability
Mean cosine to 1.0× reads
Reads ratio
DeepVelo(VAE)
DeepVelo(GCN)
UniTVelo
STT
scVelo(dyn)
SDEvelo
PhyloVelo
dynamo
velocyto
TopicVelo
biVI
veloVI
DeepKINET
cell2fate
scVelo(sto)
cellDancer
MultiVelo
TFvelo
VeloVAE
VeloAE
LatentVelo
c
Historical composite (19 configurations)
Overall Score
Accuracy rank index
Scalability rank index
Stability rank index
Ranking
Score
0%
100%
d
Docker
VeloGIF framework
Containerized RNA velocity analysis
Algorithms
scRNA-seq
velocyto
scVelo
VeloAE
Dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
STT
TFvelo
TopicVelo
biVI
DeepKINET
SDEvelo
cell2fate
Multi-omics
MultiVelo
PhyloVelo
Input
Spliced / unspliced RNA
U
S
VeloGIF
Velocity inference
Output
RNA velocity
Visualization & evaluation
Fields · transitions · scores

Fig. 6 | Computational cost, read-depth stability and the VeloGIF framework. a, Runtime across 1,000–60,000 cells, with labels rounded up to minutes or hours and colors capped at 60 minutes. Blank cells indicate unavailable measurements. **b,** Mean per-cell velocity cosine relative to full-depth predictions (1.0× reads). The 1.0× point is a self-comparison; reduced-depth points assess prediction consistency, not biological accuracy. The cell2fate results use 9,144 shared cells and 472 shared genes. TopicVelo denotes integrated-transition displacement in fixed full-depth common-gene Ms coordinates, using a single-topic stochastic configuration. **c,** Summary of 19 method configurations across accuracy, scalability and stability. Stability uses the mean signed velocity cosine at 0.2, 0.4, 0.6 and 0.8× reads against 1.0× reads. The self-comparison at 1.0× is excluded. All 19 configurations have stability results. TopicVelo uses its integrated transition matrix projected into fixed full-depth common-gene Ms coordinates. This transition-projected measure differs from the direct gene-velocity cosine used for cell2fate and other methods. Overall Score = 0.6×Accuracy rank index + 0.4×Scalability rank index; stability does not enter this score. Accuracy is aggregated over available domains among eight, and coverage/preprocessing differ among methods. Circle area maps signed cosine/GDC to (x + 1)/2 for visualization; signed values and coverage are retained in Source_data. This frozen 19-configuration composite retains eight documented endpoint differences from later sources; the new PhyloVelo diagnostic is not included. **d,** Overview of the VeloGIF framework and included methods. Method coverage varies across datasets.